Assignment 4: Multiclass SVM on Fashion-MNIST

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1 Introduction

In this assignment, I implemented a nonlinear Support Vector Machine (SVM) using the dual formulation and applied it to multiclass classification on the Fashion-MNIST dataset.

The classifier was implemented from scratch and trained using a polynomial kernel. Since Fashion-MNIST contains ten classes, I extended the binary SVM using two multiclass strategies:

- One-vs-Rest (OvR)
- One-vs-One (OvO)

I compared the performance of these approaches, tuned the regularization parameter C, and analyzed the results using confusion matrices.

2 Task 1: Nonlinear SVM

The nonlinear SVM was implemented using the dual optimization problem:

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j t_i t_j K(x_i, x_j)$$

subject to

$$0 \leq \alpha_i \leq C$$

$$\sum_{i=1}^{N} \alpha_i t_i = 0$$

where $t_i \in \{-1, 1\}$ are the class labels and $K(x_i, x_j)$ is the kernel function.

A polynomial kernel was used:

$$K(x, y) = \left(\frac{x^T y}{d} + 1 \right)^2$$

where d is the feature dimension.

After solving the quadratic program, support vectors were identified as the training points with non-zero α_i. The prediction function is

$$f(x) = \sum_{i \in SV} \alpha_i t_i K(x_i, x) + b$$

3 Task 2: Multiclass SVM (OvR vs OvO)

To handle the 10 classes in Fashion-MNIST, I implemented two multiclass strategies.

3.1 One-vs-Rest

For OvR, ten binary classifiers were trained. Each classifier distinguished one class from all other classes. During prediction, the classifier with the highest decision score determines the final label.

3.2 One-vs-One

For OvO, a classifier was trained for each pair of classes, resulting in

$$\binom{10}{2} = 45$$

binary classifiers. Prediction was performed using majority voting.

The results are shown in Figure 1. OvO achieved slightly better test accuracy than OvR.

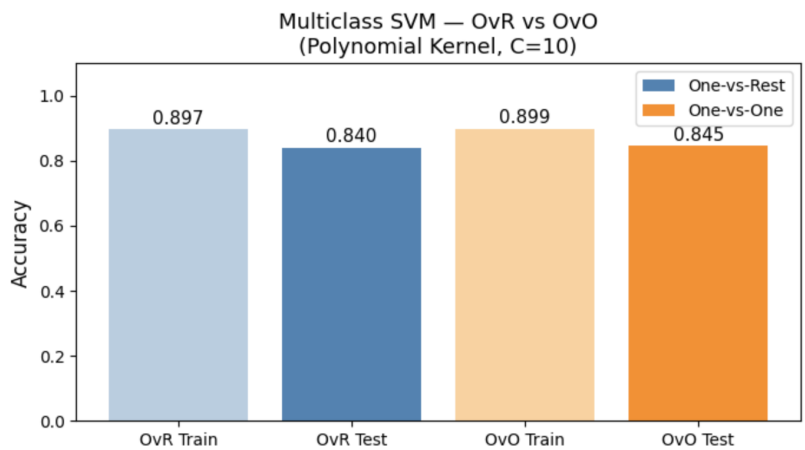


Figure 1: Comparison of OvR and OvO SVM performance.

OvR achieved a test accuracy of 0.840, while OvO achieved 0.845.

4 Task 3: Hyperparameter Tuning

The regularization parameter C was tuned using the validation set.

The candidate values were

$$C \in \{0.1, 0.5, 3, 25.12, 100, 1000\}$$

Figure 2 shows validation accuracy as a function of C.

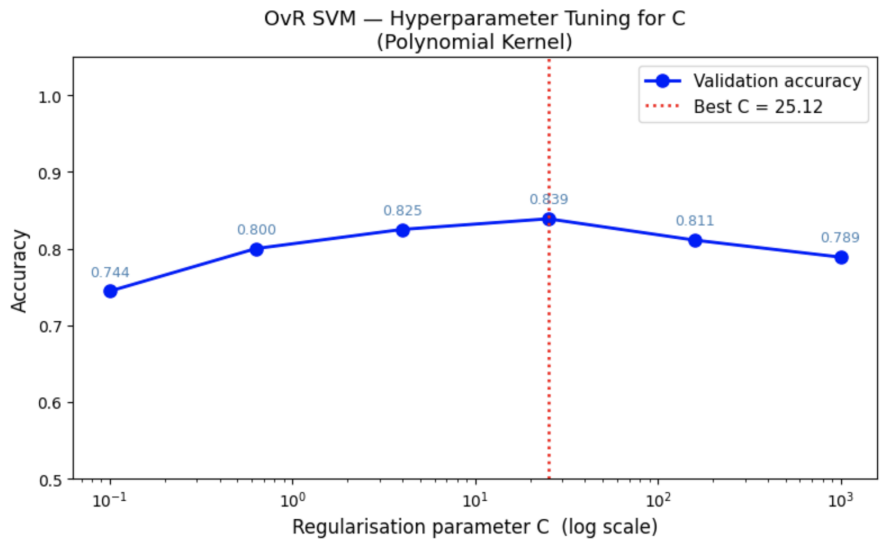


Figure 2: Validation accuracy vs regularization parameter C.

Validation accuracy increases as C increases and reaches the best performance at

$$C = 25.12$$

After this point the performance slightly decreases, indicating possible overfitting.

5 Task 4: OvR Confusion Matrix

The confusion matrix for the OvR classifier is shown in Figure 3.

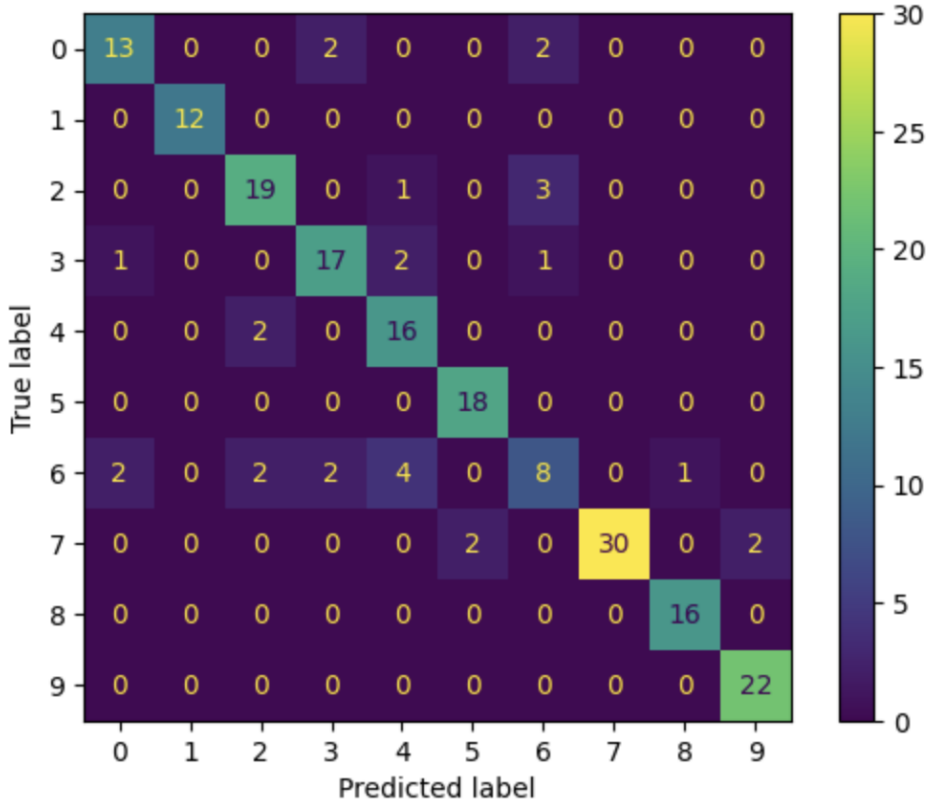
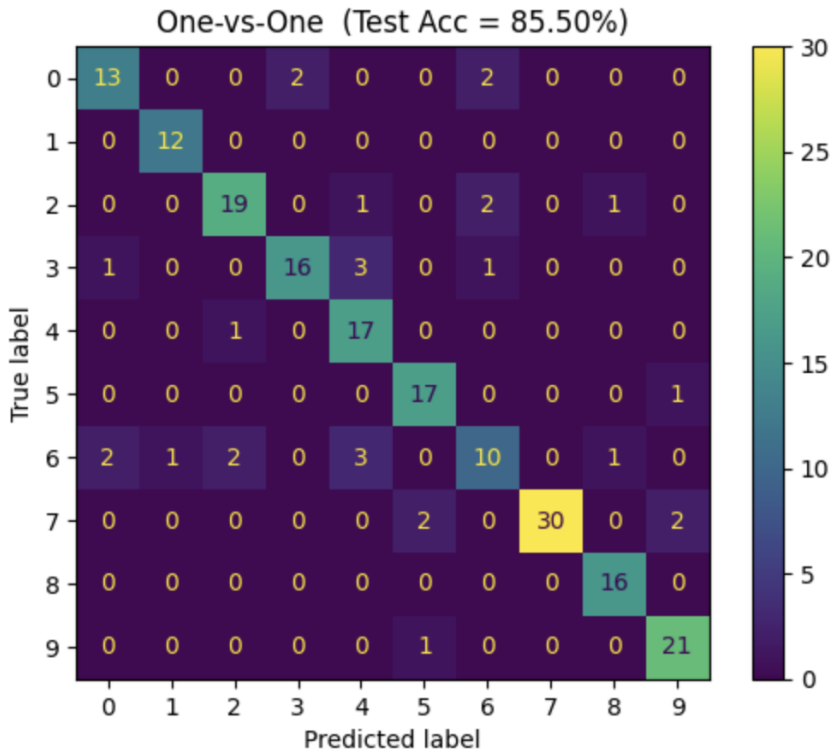


Figure 3: Confusion matrix for the OvR classifier.

The matrix shows that most predictions lie on the diagonal, indicating correct classifications. However, some confusion occurs between visually similar clothing categories such as shirts, pullovers, and coats.

6 Task 5: OvO Confusion Matrix

The confusion matrix for the OvO classifier is shown in Figure 4.

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One-vs-Rest confusion matrix plotted above (Test Acc = 85.50%)
One-vs-One  confusion matrix plotted above (Test Acc = 85.50%)
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Figure 4: Confusion matrix for the OvO classifier.

OvO achieves slightly better classification performance. Some difficult categories remain challenging, but overall the model produces accurate predictions for most classes.

7 Conclusion

In this assignment, a nonlinear SVM was implemented using the dual formulation and applied to multiclass classification of Fashion-MNIST images.

Both OvR and OvO methods achieved good performance. OvO slightly outperformed OvR in test accuracy. Hyperparameter tuning showed that the best regularization parameter was $C = 25.12$.

Confusion matrices revealed that classification errors mainly occur between visually similar clothing categories. Overall, the polynomial kernel SVM performed well on this dataset.